



ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY
COLLEGE OF ELECTRICAL AND MECHANICAL ENGINEERING
DEPARTMENT OF SOFTWARE ENGINEERING

Mobile Computing and Programming

Course Code: SWEG4103

Target Group: Software Engineering

Year : IV

Semester: II

By: Muleta Taye T.

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December 7, 2022

- What is Expected at the end ?

► What is Expected at the end ?

- Define, Explain and state the key concepts in mobile computing, emerging and possible future technologies and its application areas.
- Critically articulate, analyze and evaluate security concerns of the mobile computing technologies and mobile networks.
- Perform/conduct simulation for studying and predicting challenge and behaviors of various mobile computing technologies.
- Choose, Practice legal and Ethical principles mobile application framework for designing, and constructing mobile Software application

Assessment		
Continuous Assessment Type		Weight (in %)
1	Test	15%
2	Assignment (Seminar)	10%
3	Quiz	5%
4	Project	15%
5	Other	5%
6	Final Exam	50%
Total :		100%

Text Book and References :	1: FUNDAMENTALS OF MOBILE COMPUTING latest Edition , Kindle Edition
	2: Fundamentals of Mobile and Pervasive Computing: Frank adelstein, Sandeep K.S Gupta, Golden G. Richard (Latest Edition)
	3: Mobile Computing Principles" by ,Reza B'Far
	4: Wireless Networks and Mobile Computing,, Koushik Sinha, Sasthi C. Ghosh, Bhabani P. Sinha
	5: Mobile computing principles: Designing, Developing and Mobile Application with UML and course

Chapters

01 Fundamentals of Mobile Computing

- Introducing → Mobile Computing , Mobility,
- MC infrastructures, Application areas, Challenges in MC

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- Location Information, Security, and Privacy
- Distinguishing Privacy and Security.
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04 Mobile Programming

CHAPTER 01

Fundamentals of Mobile Computing

- What is Mobile Computing ?
- What are the Components of MC ?
- Where do we apply MC?
- Anything tough ?

CH01: Fundamentals of Mobile Computing

⇒ Is this Mobile Computing ?



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desire to move either to wards or away from resource → SCARCITY!

- There are 2 aspects of mobile computing

1. User mobility

- users communicate “anytime, anywhere, with anyone”
- user who has access to similar telecom services at different place



User moves the services follow him

2. Device Portability

- The ability a devices to be connected anytime and anywhere to the network.
- Mobility $\mapsto$ Core issue in Mobile Communication System.
- Mobile Computing is Computing and communication on the move with permanent connectivity
- The major elements or infrastructures of mobile computing are

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 - Mobile communication,
 - Mobile hardware,
 - Mobile software

CH01: Fundamentals of Mobile Computing

Mobile Hardware

- YOU KNOW!
- device components that can be used to receive or access the service of mobility
- must capable of operating in full-duplex

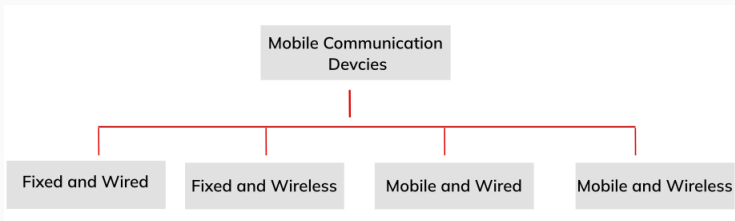
Mobile software

- is a program that runs on mobile hardware. ↳ Heart of mobile system
- This is designed to deal capably with the characteristics and requirements of mobile applications.
- This is the operating system for the appliance of mobile devices.



Mobile communication

- refers to an infrastructure that ensures seamless and reliable communication among wireless devices
- More, technical pieces that allow devices to communicate
↓
such as protocols, services, bandwidth, and portals necessary to facilitate and support the stated services.
- These devices are responsible for delivering a smooth communication process.



Mobile Communication devices categories

1. Fixed and wired

- devices are **tied at a position** , and connected through a **physical link to communicate** with other devices.
- Ex: Desktop appliances

2. Fixed and wireless

- devices are **tied at a position** , and connected through a **wireless medium to communicate** with other devices.
- Ex: Wi-Fi router

3. Mobile and Wired

- devices must be carried from place to place, reconnecting to a network via the telephone network and a modem.
- Ex: Laptops

4. Mobile and Wireless

- devices can communicate with each other irrespective of their position.
- They can also connect to any network without the use of any wired device.
- Ex: Wifi- Portable Router

★ How do MCs Implemented ? $\mapsto$

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★ How do MCs Implemented ? $\mapsto$ Reference model



Recap OSI and TCP/IP!!

CH01: Fundamentals of Mobile Computing

★ How do MCs Implemented ? $\mapsto$ Reference model



Recap OSI and TCP/IP!!

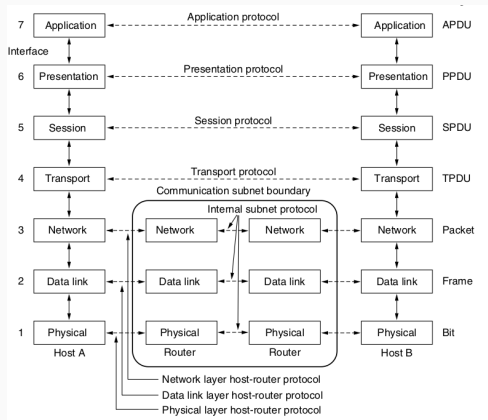


Figure 1: OSI Reference Model

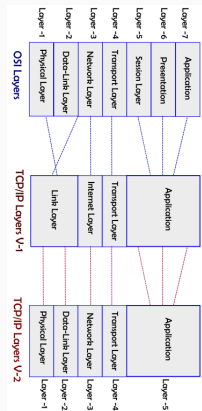


Figure 2: TCP/IP Reference Model

- MCs adopted reference model $\mapsto$ **Modification**

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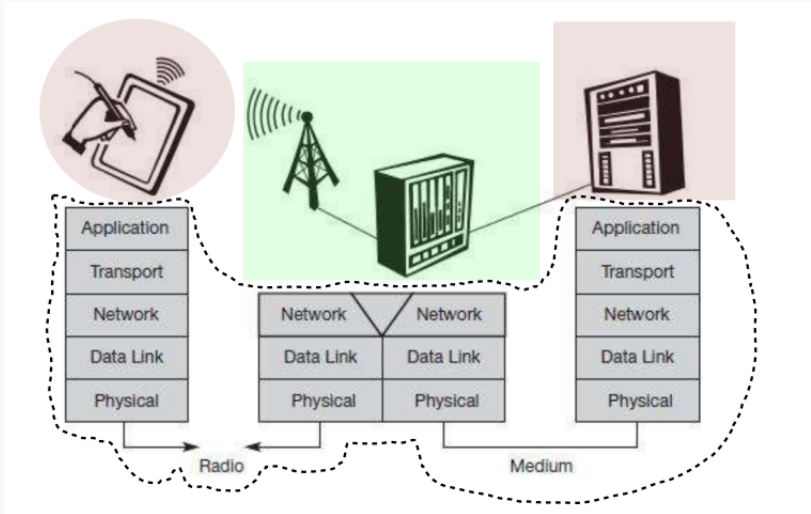


Figure 3: A Simplified Reference Model of Mobile Computing sys

Components ...

- End-systems
 - Any Mobile Computing Devices
 - Such as PDA, smart phones, Kiosk, etc
 - Devices requires protocol stack comprising the layers

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- End-systems
 - Any Mobile Computing Devices
 - Such as PDA, smart phones, Kiosk, etc
 - Devices requires protocol stack comprising the layers
- Intermediate systems $\mapsto$ Communication System
 - such as the interworking unit/devices $\mapsto$ routers or gateway devices
 - Some Layers disappeared!
 - Layers designed to handle specific action in communication.
 - A layer created with a different abstraction is needed.
 - Each layer should perform a well-defined function.
 - The function of each layer should be chosen with an eye toward defining internationally standardized protocols.
 - The layer boundaries should be chosen to minimize the information flow across the interfaces.

► Reference model

1. Physical layer:

- The bottom layer of the **Simplified reference model**
- Responsible for :

► Reference model

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- Responsible for :
 - **Encoding and signaling**
 - Sender $\mapsto$ bits into signals
 - Receiver $\mapsto$ Signal into bits
 - **Encryption**
 - Securing the computer assets
 - **Frequency selection**
 - **Generation of the carrier frequency**
 - **Modulation**

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2. Data-Link Layer

- This layer is responsible for :
 - **Accessing the medium**
 - **multiplexing of different data streams**
 - **Error Control**
 - Generally responsible for a reliable **point-to-point** OR **point to multipoint connection**

3 Network Layer

- The Third layer of SRM of Mobile Computing system



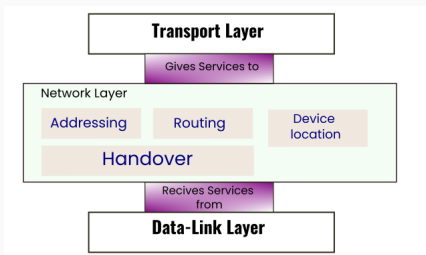
Responsible for routing packets through a network

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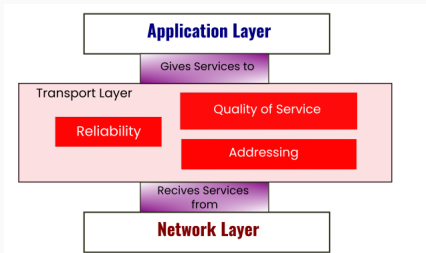
Responsible for routing packets through a network



- Packetizing
 - encapsulates packets received from upper-layer protocols
- Fragmentizing
 - decapsulates the IP datagram from the received frame, processes it, and then encapsulates it in another frame

4 Transport Layer

- The reference model to establish an end to end connection.

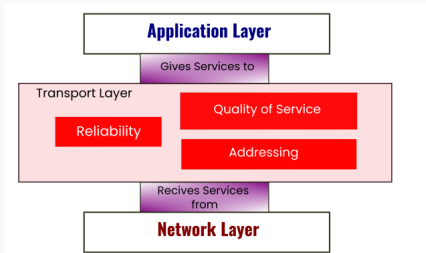


- Reliability
 - Flow control
 - Congestion control
- Quality of Service (QoS)
 - general metrics for measuring the performance of computing systems

▶ Read the detail
- Addressing →

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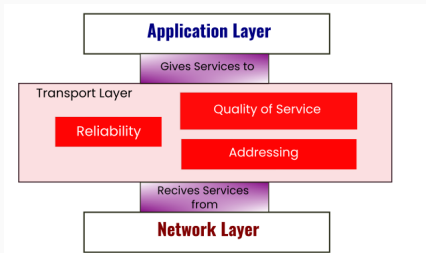
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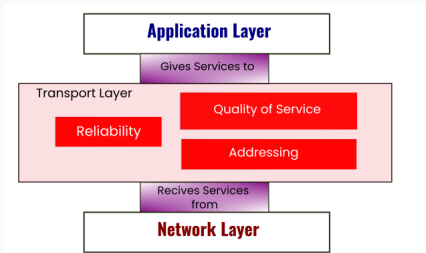


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5 Application Layer

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5 Application Layer

- review the layer

Mobile Computing Architecture

- Mobile Computing Architectures are :
 - a way of defining set of layer between disparate components of computing



User Application Interface, Device Hardware and Communication Network

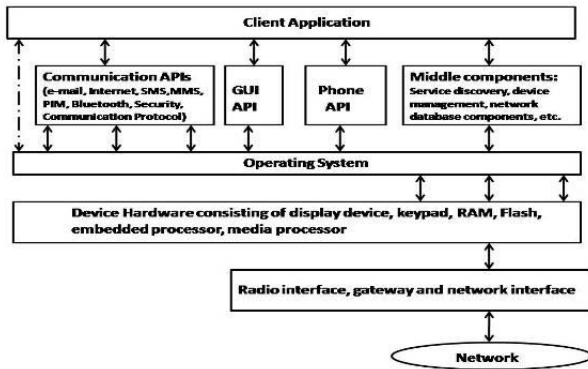
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- Mobile computing architecture uses **Three tier Architecture**

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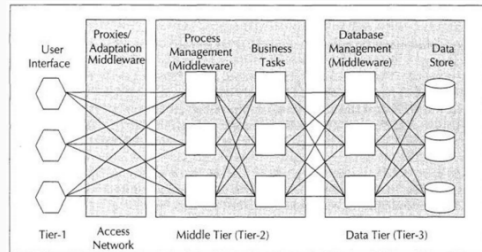
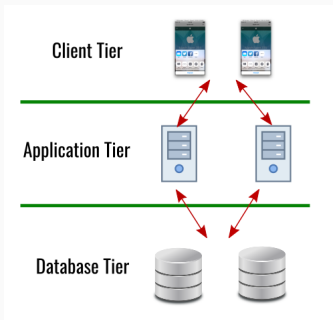


Figure 4: 3-Tier architecture of mobile computing

Presentation layer: Tier-1

- aka UI or Client layer
- Responsible for presenting the information to the end user
- moreover
 - Allows data manipulation and data entry
 - requests the data from Business layer
- Apps, browsers and customized user apps.
- It should be context-aware and device-independent

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Application layer: Tier-2

- aka Process Management layer
- Is the engine of a ubiquitous application
- Responsible for processing user input, obtaining data and making decision.
- How does Tier-2 functions Implemented.



Implemented using Middleware software

♣ Types of Middleware

- Message Oriented Middleware (MOM)
 - designed to connects different applications through exchange of messages.
 - message queue is used to store the messages until it can be processed.
- Transaction Processing (TP) Middleware
 - Provides a mechanism for developing transaction-based distributed applications.
 - It includes management features like restarting failed processes and enforcing consistency of distributed data.
 - client requests $\mapsto$ different application tasks
- Communication Middleware – used to connect one application to another.
- Transcoding Middleware
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Data Access Layer (DA)

- devised to **store data** needed by the application.
- Database middleware runs between **the application program** and **the database**
- Ex: any RDBMS

► Mobility Vs IP

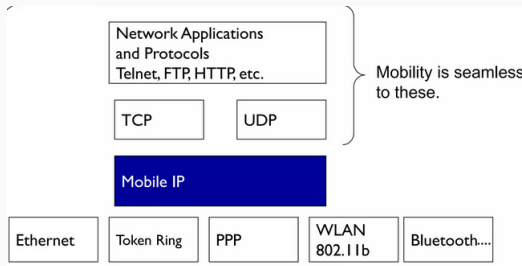
- Recall Mobility!
 - You Know IP!
 - 2-Kind of Mobility with respect to applications
 1. Seamless Mobility : $\mapsto$ **check thoroughly**
 2. Portable Mobility
 - mobility is not transparent to applications. but transmission is possible!
- ↓
- How ? $\mapsto$ When a device change the point of attachment?**

- Mobile IP

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- Mobile IP

- is IETF (Internet Eng. Task Force) standard protocol
 - "Designed to allow mobile device users to move from one network to another while maintaining permanent IP"
- Layer-3 (i.e network layer)
- Compatible with Higher-Layer and Lower-Layer protocol



- 2-type of address Mobile IP uses
 - **Home Address** $\mapsto$ For identifying the devices
 - **Care-Of-Address** $\mapsto$ For the Purpose of Routing

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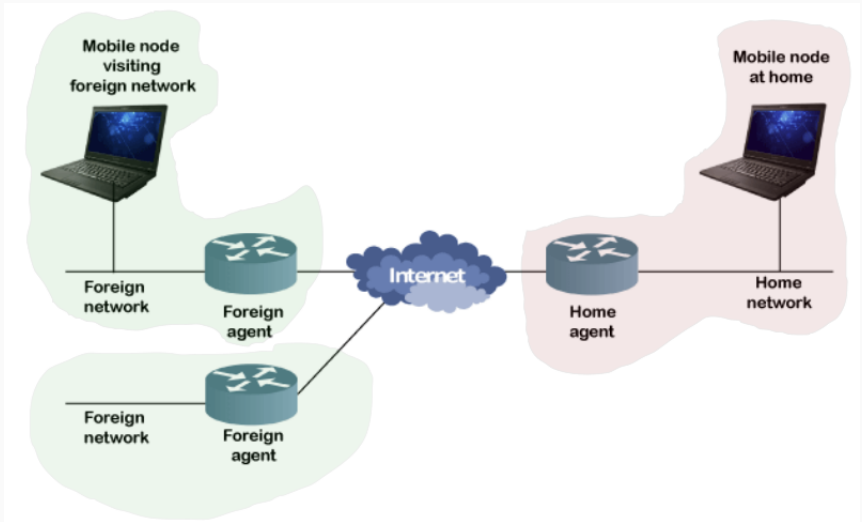


Figure 5: Mobile IP Architecture

Components of Mobile IP

1. Mobile Nodes

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5. Foreign Agent

- is a default router in the foreign network
- Provide **several services** to the mobile node during **its visit to the foreign network**.

- Care-Of-Address $\mapsto$

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- Care-Of-Address $\mapsto$ Critical!
 - is IP-Address of the node when Operating in foreign network!
 - Identify
 - Current Location of Nodes
 - Sent by foreign agent to Home Agent
 - Ip-address of foreign agent
 - It is used to deliver data packets through the process of tunneling.
 - Packets to Mobile node sent to $\mapsto$

CH01: Fundamentals of Mobile Computing

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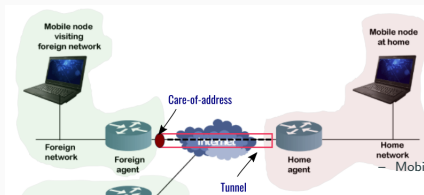
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it can act as its own foreign agent
 - DHCP is used to assign IP Address incase required.

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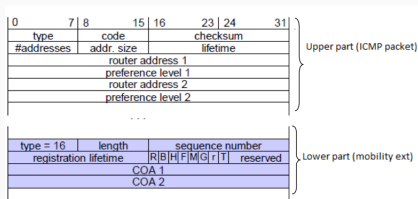
- The mobile IP process has three main phases :

1. Phase - 1 $\mapsto$ Agent Discovery

- is the method by which **a mobile node determines where they reside.**
- then **Home Agent** and **Foreign Agents** advertise their presence via
- Mobile IP extends **ICMP router discovery** $\mapsto$ For agent discovery
- Two methods:
 - **Agent Advertisement**
 - FA and HA advertise their presence periodically using special **agent advertisement messages.**
 - ICMP is employed
 - **Agent solicitation**
 - **what if no advertisement present or arrival time is High?**



Mobility agents sends **Agent solicitation** beacon packet



1. Phase - 2 $\mapsto$ Registration

- "provides a flexible mechanism for mobile nodes to communicate their current reachability information to their home agent."

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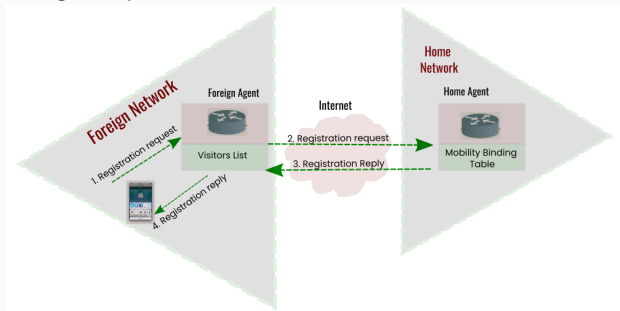
$\downarrow$ means

Inform the home agent of the current location for correct forwarding of packets

- carried out in 2 antithetic ways

1.1 Via mobility binding $\mapsto$ If the COA is at the FA

- The Node sends its registration request containing the COA to the FA $\mapsto$ forwarding the request to the HA



- Mobility Binding Table

HOME ADDRESS	Care-of-Address	TTL
192.168.10.12 192.168.10.14	10.10.10.34 120.120.10.34	200 sec 120 sec

- Visitor List Table

Home address	HA address	Media address	TTL
192.168.10.12 162.16.10.14	192.168.10.1 162.16.10.1	00-00-00-68-AF-F1 05-70-00-68-A6-C1	200 sec 120 sec

CH01: Fundamentals of Mobile Computing

- Registration ...

- 2 If the COA is co-located

- The mobile node may send the request directly to the HA and vice versa

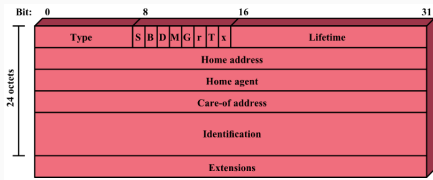
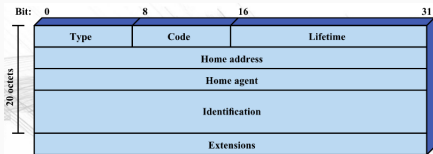


Figure 6: Registration Reply request Format



3 Phase -3 $\mapsto$ Tunneling

- **Port-Forwarding** $\mapsto$ The method sending IP datagram
- establish a virtual pipe for data packets between a tunnel entry and a tunnel endpoint.
- Packets which are entering in a tunnel are forwarded inside the tunnel and leave the tunnel unchanged.
- IP datagrams flow
 - **Source $\mapsto$ Encapsulator $\mapsto$ Decapsulators $\mapsto$ Destination**
- HA intercept the packet and **Encapsulate it** and tunnels to COA of the mobile
 - Encapsulated bundle delivered to COA using routing
 - Encapsulation : **payload with an additional header**
 - 3-Kind of encapsulation
 - **IP-IP Encapsulation**
 - **Minimal Encapsulation**
 - **Generic Routing Encapsulation**

Encapsulation

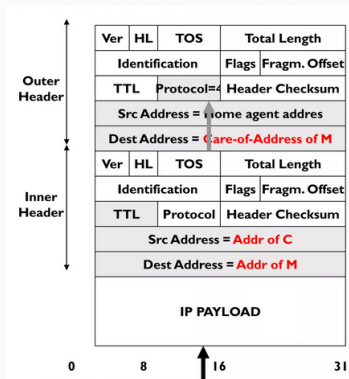
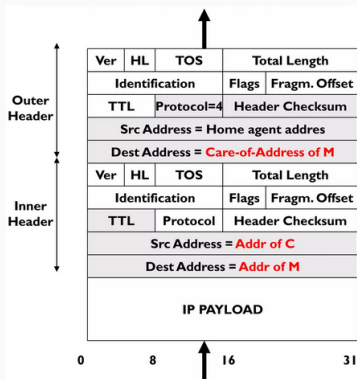
- **IP-IP Encapsulation**
 - Entire IP datagram becomes payload in new IP datagram
 - Original, inner IP header unchanged except TTL decremented by 1

CH01: Fundamentals of Mobile Computing

Encapsulation

• IP-IP Encapsulation

- Entire IP datagram becomes payload in new IP datagram
- Original, inner IP header unchanged except TTL decremented by 1



- Encapsulated pkt received @ FA
- Receiver understand knows it is IP-IP encapsulation by protocol = 4

- Design Consideration in MCs ?
 - Core $\mapsto$

- Design Consideration in MCs ?

- Core $\mapsto$ Contextualization



- Design Consideration in MCs ?

- Core $\mapsto$ Contextualization



- Context-independent and Context-sensitive in terms of content and behavior
- Context can be used as a basis by which an adaptation manager or algorithm decides to modify content or application behavior.
- Different types of context
 - Client Context Manager
 - Policy Manager
 - Adaptability Manager
 - Security Manager

Context ...

- Client Context Manager
 - It gathers and maintains information related to the client device, user, network, and environment surrounding the mobile device.

↓ How?

Context ...

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 - ↓ How?
 - via set of Awareness Modules
 - Context-aware systems $\mapsto$ Cool!

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↓ How?

- via set of Awareness Modules

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↓

Capturing, Representing, Filtering, and Interpreting contextual data.

- Managed fro three different aspects (i.e. Context Management)

- Context sensing,
- Context representation,
- Context interpretation.

- Policy Manager

- is a set of rules.
- are responsible for controlling policies related to mobility.
- A policy in combination with role and current context information can determine what actions a user is allowed to perform.

Context ...

- Security Manager
 - Provides **secure connection** between the client device and the server.
 - Depending on the security policies,



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- It ensures security with respect to
 - Confidentiality $\mapsto$ **Encryption**
 - Integrity $\mapsto$ **Hashing**.
 - Availability $\mapsto$ **Firewall**.
 - Non-repudiation $\mapsto$ **Digital signature**.
 - Trust $\mapsto$ **Third-party recommendation system**
- Adaptability Manager
 - Is responsible for **adapting content** and **behavior** according to context and policy.
 - **Content Transcoding** to be viewed on a particular device

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- Adaptability Manager

- Is responsible for **adapting content** and **behavior** according to context and policy.
 - **Content Transcoding** to be viewed on a particular device
- Content Adaptation needs:
 - **Physical capabilities of the device** $\mapsto$ screen size, width/height. . . etc.
 - **Logical capabilities of the device** $\mapsto$ displaying video. . . etc.
 - **Effective network bandwidth**
 - **Payload**

CH01: Fundamentals of Mobile Computing

- Any Challenges in Mobile Computing system?

- **Any Challenges in Mobile Computing system?** $\mapsto$ unquestionably!
 - Categorized into three major areas:
 1. **Lack of standards**
 2. **Connectivity**
 3. **Mobility**
 4. **Portability**

Challenges in Mobile Computing

- **Lack of standards**
 - lack of pre-written spec/ standards
- **Connectivity**
 - **Disconnection** $\mapsto$ more prone to disconnection.
 - **Scarcity of Bandwidth** $\mapsto$ divided in to available cell.
 - **Heterogeneous Network** $\mapsto$ nodes confront multiple network
 - **Security** $\mapsto$ The nature of the Link
- **Portability**
 - Consumes power consumption
 - pragmatic challenges
 - Chance of data loss, small user-interface issues
 - Systems issues
 - network integration, resource imbalance such as lack of local storage.

♣ Any Usage ?

♣ Any Usage ? $\mapsto$ YUP!

♣ Any Usage ? $\mapsto$ YUP!

- Vehicles
- Emergencies
- Credit Card Verification
- Replacement of Wired Networks
- Infotainment
- Location dependent services

Review the application of Mobile computing?



Done!
Question!